

# Software Requirements Specification (SRS)

## SmartFarm Management Application

|                       |                                                  |                           |
|-----------------------|--------------------------------------------------|---------------------------|
| Document Version: 1.0 | SDLC Stage: Phase 1 — Requirements Specification | Status: Approved Baseline |
|-----------------------|--------------------------------------------------|---------------------------|

### 1. Introduction

#### 1.1 Purpose

This Software Requirements Specification (SRS) document details the complete functional, non-functional, security, and architectural requirements for the **SmartFarm Management Application**. It serves as the authoritative single source of truth for engineering, testing, and continuous maintenance.

#### 1.2 Scope of the System

SmartFarm is an enterprise-grade, zero-cost cloud farm management platform designed for agricultural enterprises. The application enables:

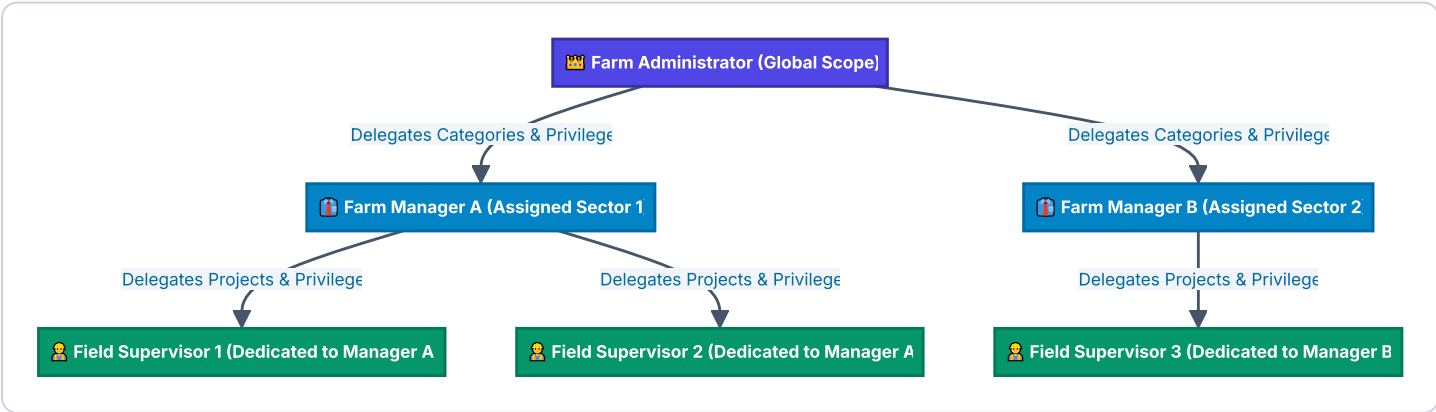
- Executive oversight and financial tracking by **Farm Administrators**.
- Sector-level resource management, project planning, and supervisor delegation by **Farm Managers**.
- Field-level yield, activity, and input tracking by **Field Supervisors** under strict financial privacy isolation.

#### 1.3 Definitions, Acronyms, and Abbreviations

- RBAC:** Role-Based Access Control.
- PBAC:** Privilege/Policy-Based Access Control (fine-grained feature toggle permissions).
- 1:N:** One-to-Many entity relationship.
- Zero-State Dashboard:** A tailored, clean UI rendered when an authenticated user has zero assigned projects/categories, suppressing empty graphs and displaying actionable next steps.
- KPI:** Key Performance Indicator (Revenue, Yield, Expenses, Net Profit).

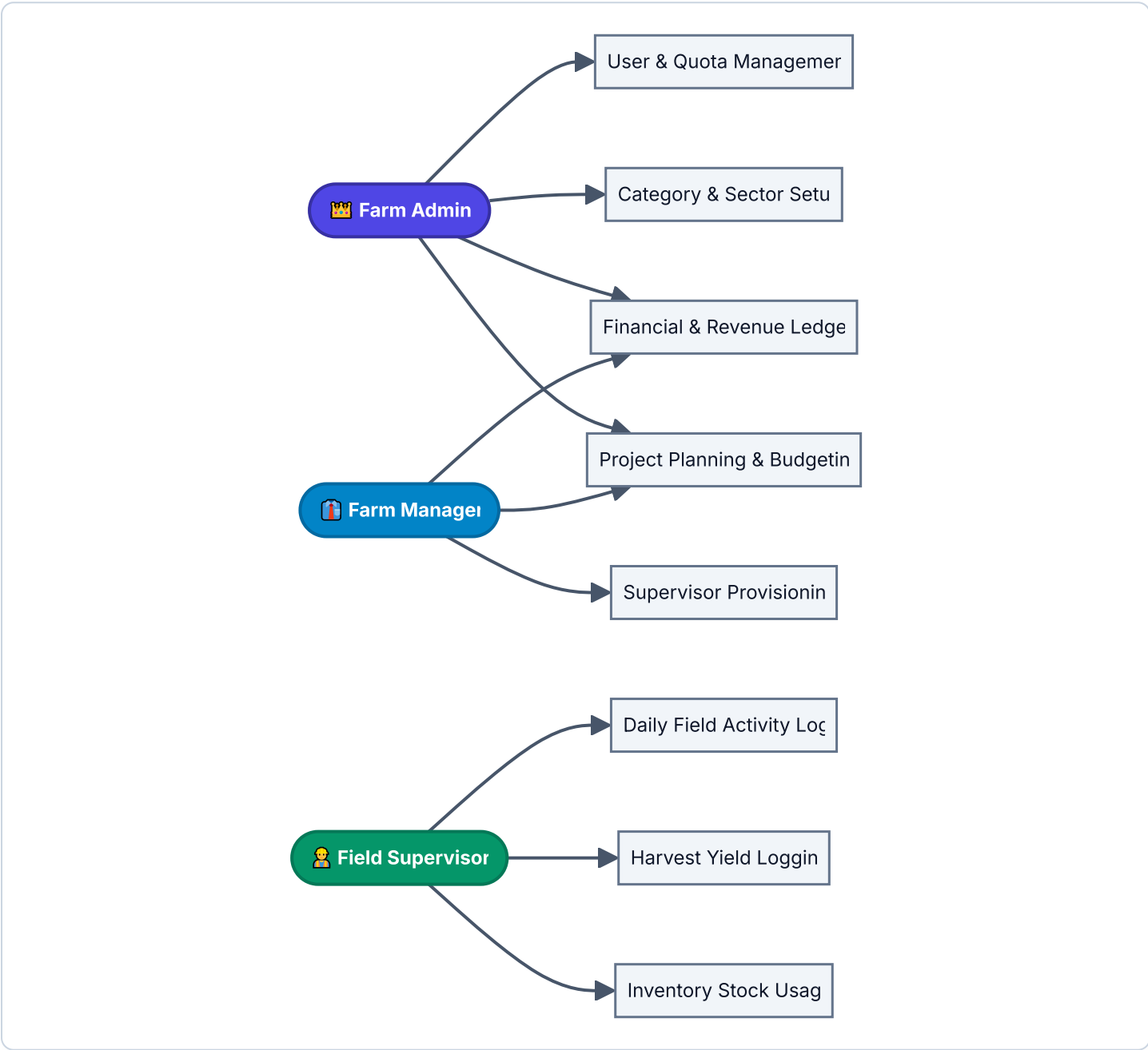
### 2. Overall Description & System Context

#### 2.1 Three-Tier Organizational Hierarchy



| Persona                | Role Key     | System Scope         | Max Quota       | Core Responsibilities                                                                                       |
|------------------------|--------------|----------------------|-----------------|-------------------------------------------------------------------------------------------------------------|
| **Farm Administrator** | `ADMIN`      | Entire Farm (Global) | 1 Account       | User management, financial health, category assignment, manager privilege delegation, system configuration. |
| **Farm Manager**       | `MANAGER`    | Assigned Categories  | Max 2 Accounts  | Project planning, supervisor hiring & assignment, sector expense auditing, supervisor privilege delegation. |
| **Field Supervisor**   | `SUPERVISOR` | Assigned Projects    | Max 10 Accounts | Daily field logging (harvests, activities, stock consumption). **Zero access to sales revenue/budgets.**    |

2.2 System Use Case Overview



3. Detailed Functional Requirements (FR)

FR-1: Authentication, Account Provisioning & Password Recovery

FR-1.1: System shall support authentication using either `Username` or `Email` combined with a password.

**FR-1.2:** Unauthenticated users can request a password reset by providing their registered email address ( `POST /users/forgot-password` ).

**FR-1.3:** If an email is not registered in the system, the API shall return HTTP 400 with: `"No account found with this email address. If you did not register an email, please contact your Farm Administrator to reset your password."`

**FR-1.4:** If the email exists, the system generates a secure cryptographic token ( `PasswordResetToken` valid for 1 hour) and sends an HTML reset email via SMTP.

**FR-1.5:** Administrators and Managers can execute direct password resets for staff accounts via the User Management portal ( `PATCH /users/{id}/admin-reset-password` ).

## FR-2: Staff Quotas & Hierarchical Access Delegation (PBAC)

**FR-2.1:** The system shall strictly enforce organizational staff limits: Max 1 Admin, Max 2 Managers, Max 10 Supervisors.

### FR-2.2: Admin to Manager Delegation:

- Admin assigns one or more Farm Categories to a Manager.
- Admin configures fine-grained privilege toggles for the Manager:
- `CAN_CREATE_CATEGORIES` : Ability to create new farm sectors.
- `CAN_CREATE_SUPERVISORS` : Ability to provision supervisor accounts.
- `CAN_VIEW_FINANCIALS` : Ability to view profit/loss and revenue totals.
- `CAN_MANAGE_BUDGETS` : Ability to set and modify project budgets.

### FR-2.3: Manager to Supervisor Delegation:

- Manager provisions supervisors linked to their manager account ID ( `manager_id` ).
- Manager assigns specific projects (within the Manager's categories) to the Supervisor.
- Manager configures fine-grained privilege toggles for the Supervisor:
- `CAN_RECORD_HARVEST` : Log harvest yield volumes and units.
- `CAN_LOG_ACTIVITIES` : Submit field activity and labor logs.
- `CAN_USE_INVENTORY` : Deduct fertilizer, feed, and tools from stock.
- `CAN_RECORD_EXPENSES` : Log petty field cash expenses.
- `CAN_RECORD_SALES` : Record immediate farm-gate sales without viewing global finances.

## FR-3: Workload & Capacity Validation Rules

**FR-3.1 (Manager Workload):** When assigning a category to a Manager, the system shall evaluate existing assignments. If a manager exceeds 3 active categories, an advisory workload confirmation is triggered.

**FR-3.2 (Supervisor Workload):** Each supervisor has a `max_project_capacity` (default 4 projects). The system blocks assigning additional active projects unless capacity is adjusted or existing projects are completed.

## FR-4: Category & Project Lifecycle Management

**FR-4.1:** Projects must belong to an existing Category and possess a globally unique project name.

**FR-4.2 (Date Validation Rule):** The system shall strictly validate that `startDate ≤ endDate` . If `startDate > endDate` , both backend and frontend reject the request with: `"Start date cannot be greater than end date!"`

**FR-4.3 (Unique ID Generation):** Project, Category, and Transaction IDs are generated using an incremental collision-safe loop ( `while (repo.existsById(id)) count++;` ) preventing primary key collisions after deletions.

## FR-5: Operational Field Tracking (Harvest, Activity, Inventory)

**FR-5.1:** Harvest records capture item name, numerical yield, unit of measurement (KG, Litres, Bags), date, and notes.

**FR-5.2:** Activities capture activity title, type (Weeding, Irrigation, Spraying, Feeding), date, and notes.

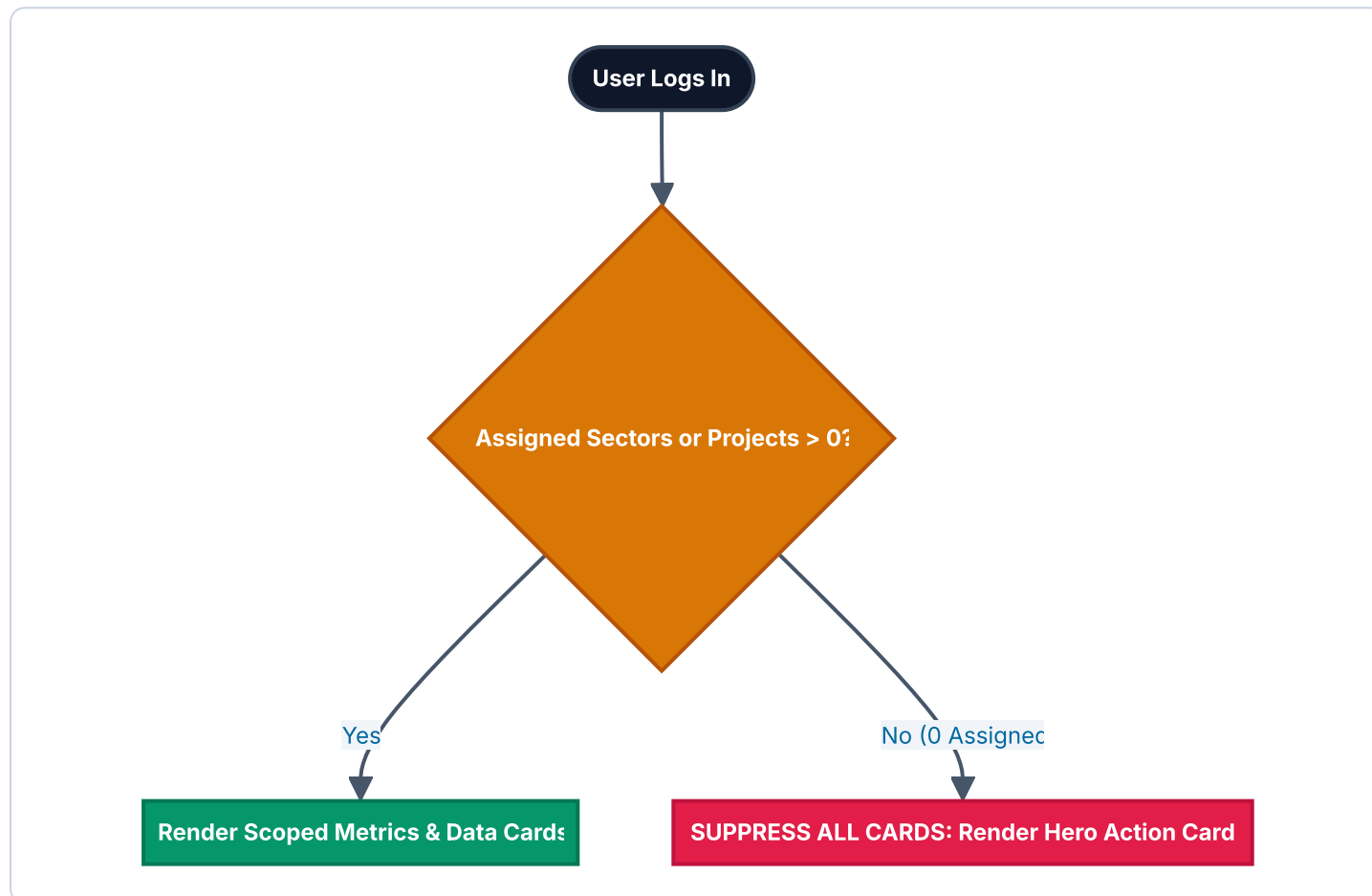
**FR-5.3:** Inventory tracks quantity in stock, unit prices, reorder levels, and records deduction history when consumed by projects.

## FR-6: Financial Governance & Sales Enforcement

**FR-6.1 (Stock-Backed Sales Rule):** A project cannot record a sale exceeding its total recorded harvest yields minus previous sales.

**FR-6.2 (Supervisor Financial Shield):** Supervisor API endpoints and dashboard views strip budget allocations, sales revenue, and profit calculations.

## FR-7: Dynamic Scoped Dashboards & Zero-State Engine



**FR-7.1 (Admin Dashboard):** Renders enterprise-wide aggregates, category financial distribution, user quotas, and global cashflow.

**FR-7.2 (Manager Dashboard):** Aggregates budget, project counts, and supervisor workloads strictly for the manager's assigned categories.

**FR-7.3 (Supervisor Dashboard):** Aggregates harvest volumes, pending tasks, and recent logs strictly for assigned projects.

**FR-7.4 (Zero-State Suppression):** If a user has 0 assigned categories or 0 assigned projects:

- All metrics cards, financial figures, and graphs are **suppressed**.
- Renders a dedicated **Action Hero Card** providing immediate contextual instructions (e.g., *"No categories assigned yet. Contact your Farm Administrator to get started."*).

## FR-8: User Deletion & Referential Integrity Safeguards

**FR-8.1:** Admin accounts cannot be deleted ( `isUserAdmin` protection).

**FR-8.2:** Deleting a user safely cascades/unlinks child records: deletes active `PasswordResetToken` rows, clears category join tables, unlinks `supervisor_id` on projects, and clears `createdById` references.

## 4. Non-Functional Requirements (NFR)

---

**NFR-1 (Security):** BCrypt password hashing (\$cost = 10\$), role-based endpoint security, parameter sanitization, and SQL injection prevention via Spring Data JPA.

**NFR-2 (Performance):** Page load times \$< 1.5s\$ on broadband; database indexed on `username`, `email`, `category_id`, `supervisor_id`, `manager_id`.

**NFR-3 (Reliability):** 99.9% uptime target backed by automated heartbeat monitoring (UptimeRobot).

**NFR-4 (Cost):** 100% compliant with zero-cost cloud architecture limits (Render Free Web Service, Vercel Hobby, TiDB Cloud / Aiven Free Tier).

**NFR-5 (Usability):** Typography optimized with `Inter` and `DM Sans`, accessible contrast ratios, and responsive mobile-first views.